

	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering	
ASSIGNMENT – 6		

Build Job using Jenkins that compiles a Java project and configures triggers to run the build on every commit.

Objective

To configure a Continuous Integration (CI) job in Jenkins that automatically compiles a Java project whenever changes are committed to a Git repository.

Tools Required

- Jenkins
- Git
- GitHub
- Apache Maven (or any Java build tool)
- Java JDK (installed and configured)

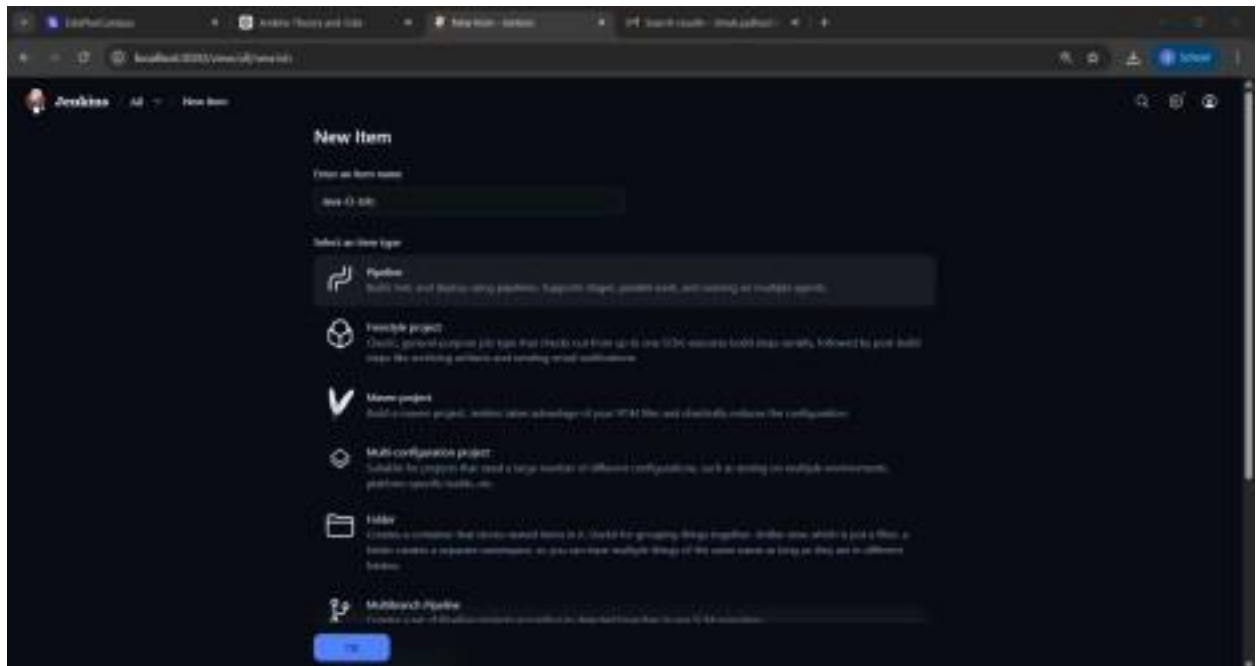
Procedure

Step 1: Start Jenkins <http://localhost:8080->

Login to Jenkins dashboard.

Step 2: Create New Job

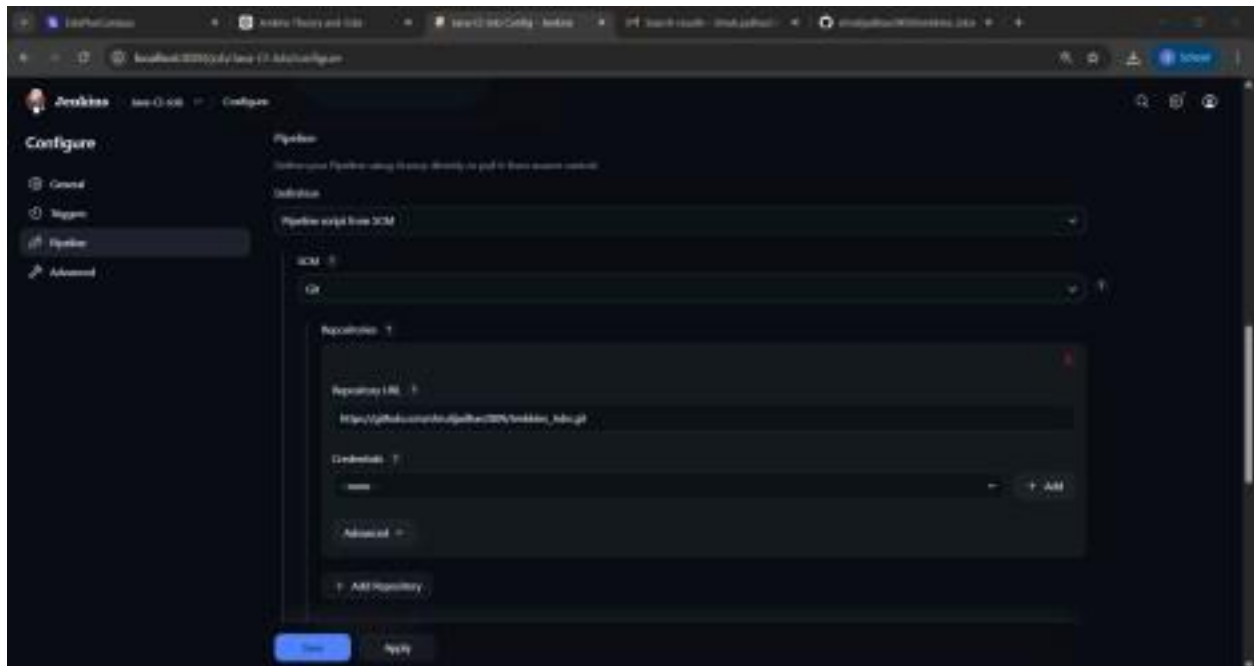
1. Click on **New Item**.
2. Enter job name (e.g., Java-CI-Job).
3. Select **Freestyle Project** or CI CD Pipeline
4. Click **OK**.



Step 3: Configure Source Code Management

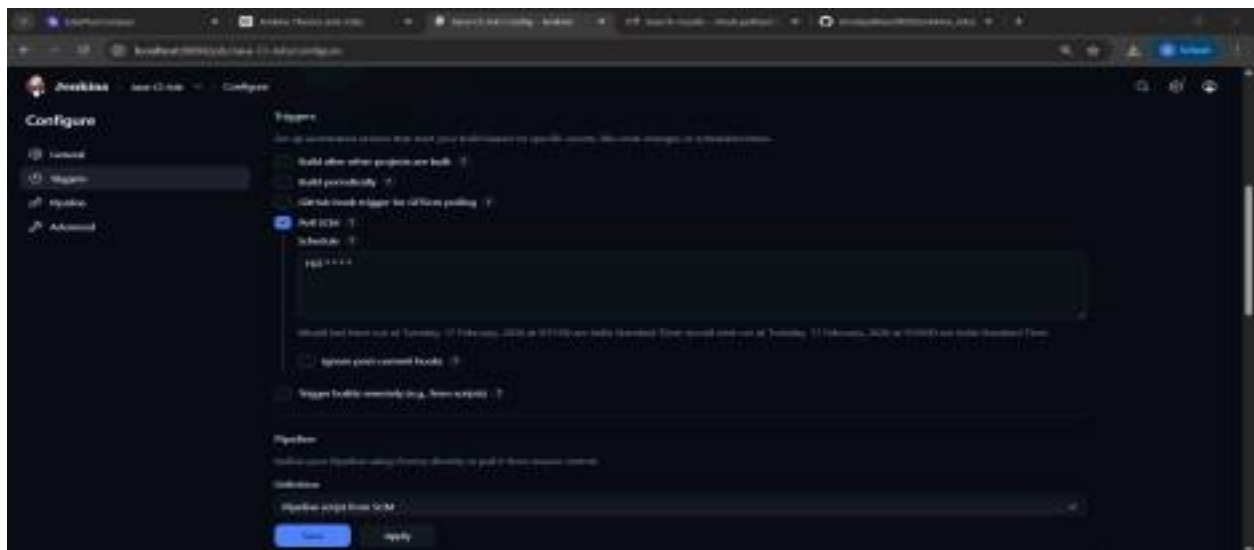
1. Go to **Source Code Management** section.
2. Select **Git**.
3. Enter GitHub repository URL.
4. Add credentials if required

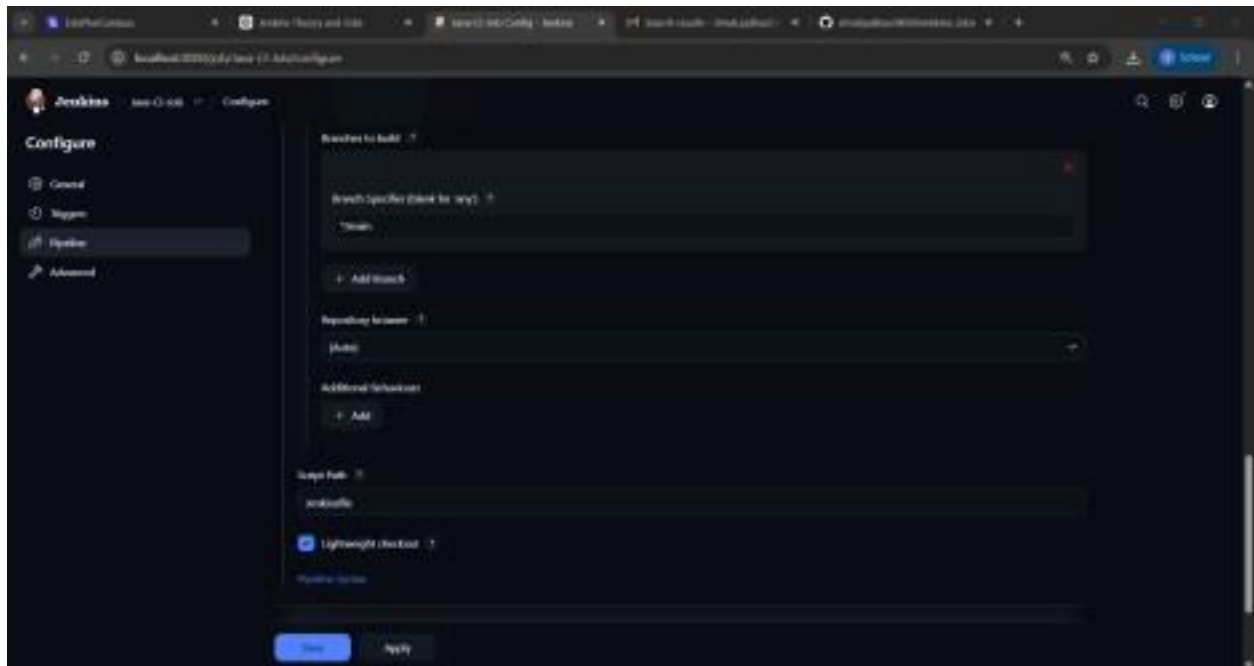




Step 4: Configure Build Trigger

1. Go to **Build Triggers** section.
2. Select **Poll SCM**.
3. Enter schedule: **(H/2 * * * *)**

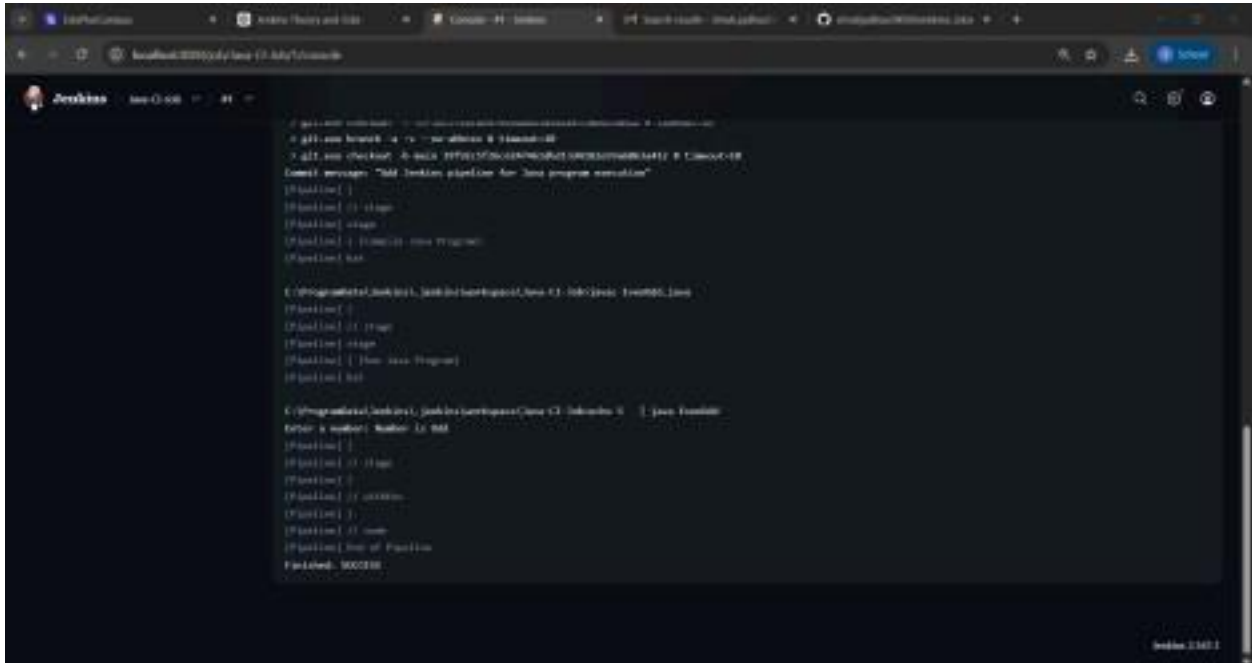




Step 5: Save and Build

1. Click **Save**.
2. Click **Build Now**.
3. Check **Console Output** to verify compilation.



A screenshot of a web browser displaying the Jenkins console output for a build. The browser has several tabs open, including 'Jenkins', 'Android Studio', and 'GitHub'. The console output shows a series of steps: 'git clone', 'git checkout', 'git pull', 'mvn compile', 'mvn package', 'mvn install', 'mvn test', 'mvn deploy', 'mvn clean', 'mvn compile', 'mvn package', 'mvn install', 'mvn test', 'mvn deploy', 'mvn clean', 'mvn compile', 'mvn package', 'mvn install', 'mvn test', 'mvn deploy'. The output ends with 'Finished: SUCCESS'.

FAQs

1. How does Jenkins automatically trigger a build when a new commit is made to the GitHub repository?

Jenkins continuously monitors the connected GitHub repository for any changes. When a developer commits new code, Jenkins detects the update through the configured source code management settings. Once a change is detected, Jenkins automatically starts the build process and compiles the Java project without any manual intervention. This ensures that the latest code is always tested and verified.

2. Why is the Poll SCM trigger used in Jenkins, and what is the purpose of the schedule H/2 * * * *?

The Poll SCM trigger is used to check the Git repository at regular intervals for new changes. The schedule H/2 * * * * means Jenkins checks the repository every 2 minutes. If any new commit is found during this check, Jenkins triggers the build automatically. This helps in maintaining continuous integration by ensuring quick feedback on code changes.

3. What role do tools like Git, GitHub, and Maven play in building a Continuous Integration (CI) pipeline for a Java project?

Git is used for version control to track code changes. GitHub hosts the Git repository and allows collaboration among developers. Maven is used as a build automation tool to compile the Java project and manage dependencies. Together, these tools help Jenkins automate the process of fetching code, building the project, and verifying changes efficiently.

Conclusion

This assignment successfully demonstrates the configuration of a Continuous Integration job in Jenkins to automatically compile a Java project whenever changes are committed to the GitHub repository. By integrating Git, GitHub, and Maven with Jenkins build triggers, the CI process ensures timely builds, early error detection, and efficient automation.